

# The cognitive interview

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## 19. THE COGNITIVE INTERVIEW

### **Introduction**

The Cognitive Interview (or CI) is one of the most exciting developments in forensic psychology in the last twenty years. The CI is a method that comprises a series of memory retrieval techniques designed to increase the amount of information that can be obtained from an eyewitness. It can therefore help investigators obtain more complete and accurate reports from interviewees. The CI was initially developed by the psychologists Ed Geiselman (University of California, Los Angeles) and Ron Fisher (Florida International University) in 1984 as a response to the many requests they received from police officers and legal professionals for a method of improving witness interviews. It is based upon known psychological principles of remembering and retrieval of information from memory. Police detectives trained to use this technique enabled witnesses to produce over 40% more valid information than detectives using their traditional interviewing techniques (see Koehnken, Milne, Memon & Bull, 1999, for a meta-analysis of research). Furthermore, university students using this new procedure obtained more information from witnesses than did experienced police officers that interviewed in their normal way (see Fisher & Geiselman, 1992; Koehnken et al, 1999; Memon & Koehnken, 1992; Memon & Higham, 1999, for reviews).

To date some 65 studies have been conducted. This includes two studies conducted in the field using real life witnesses and police officers trained in the CI technique. This chapter will provide a critical review of research on the CI and will highlight methodological and theoretical issues. Practical issues and implications for future research will be considered. Before describing the procedure and reviewing the empirical research, it is useful to understand *why* a procedure such as the CI is necessary (see also Wells et al., 2000).

### **What Led to the Development of the Cognitive Interview?**

The ability to obtain full and accurate information is critical in an investigation - it may determine whether or not a case is solved, yet the eyewitness literature reveals that such recall is difficult to achieve because of the cognitive and social factors that can influence recall accuracy (for reviews see Bruck & Ceci, 1999; Memon, Vrij & Bull, 2003). The Cognitive Interview arose out of a need to examine ways of improving witness memory (cognitive factors) as well as addressing some of the social factors that can impact on the quality of the report obtained from a witness.

Analysis of the techniques used by untrained police officers in Florida (Fisher et al., 1987) suggested that there existed some fundamental problems in the conduct of interviews that were leading to ineffective communication and poor memory performance. Fisher et al. (1987) documented several characteristics of the 'standard police interview' among which were constant interruptions (when an eyewitness was giving an account), excessive use of question-answer format and inappropriate

sequencing of questions. George (1991) studied the techniques typically used by untrained officers in London and found a remarkably similar pattern among that group. This led to the characterisation of a 'standard police interview' as being one of poor quality and stressed the need for an alternative procedure for interviewing witnesses (see Kebbell, Milne & Wagstaff, 2001, for a more recent analysis of the standard interview). In recent years, guidelines have been set out detailing the procedures that should be followed by professionals who interview witnesses (Home Office, 2003; Scottish Executive, 2003; Wells et al., 2000). However, the focus has tended to be on child interviews while there is much room for improving the quality of investigative interviews with adult witnesses (see Cherryman & Bull, 1995; Clark & Milne, 2001; Gudjonsson, 2003).

The CI represents the alliance of two fields of study: Cognition and Communication. The original version drew heavily upon what psychologists know about the way in which we remember things. Revisions of the procedure focused more heavily on the practical considerations for managing a social interaction and this was led by a desire to improve communication in police interviews and alleviate some of the problems described above. Obviously the 'cognitive' and 'communication' components work in tandem (see also Memon & Stevenage, 1996, Memon & Higham, 1999). However, for the purposes of describing the procedure as it has been depicted in the published literature, the 'cognitive' and 'communication' components will be outlined separately.

## What is the Cognitive Interview?

### The original procedure: the 'cognitive' components

The 'cognitive' part of the CI relies upon two theoretical principles. Firstly, that a retrieval cue is effective to the extent that there is an overlap between the encoded information and the retrieval cue (Flexser & Tulving, 1978) and that reinstatement of the original encoding context increases the accessibility of stored information (Tulving & Thomson's Encoding Specificity Hypothesis, 1973). The second theoretical perspective that influenced the development of the CI was the Multiple Trace Theory (Bower, 1967). This suggests that rather than having memories of discrete and unconnected incidents, our memories are made up of a network of associations and consequently, there are several means by which a memory could be cued. It follows from this that information not accessible with one technique may be accessible with another (Tulving, 1974).

### Context Reinstatement

The first technique is for the interviewee to mentally reconstruct the physical and personal contexts that existed at the time of the crime. Although this is not an easy task the interviewer can help witnesses by asking them to form an *image* or *impression* of the environmental aspects of the original scene (e.g. the location of objects in a room), to comment on their emotional reactions and feelings (surprise, anger, etc.) at the time, and to describe any sounds, smells and physical conditions (hot, humid, smoky, etc.) that

were present. Geiselman and his colleagues (Saywitz et al. 1992) have suggested that it may be helpful for child witnesses to verbalise out aloud when mentally reinstating context. For example, to describe the room as the picture comes to mind, to describe smells, sounds and other features of the context. The following is an example of how the instructions to reinstate context were administered in a study where adult witnesses were interviewed about a photography session (Memon et al. 1997a):

Interviewer: *First of all I'd like you to think back to that day. Picture the room in your head as if you were back there ... Can you see it? (pause for reply) Think about who was there (pause) How you were feeling (pause) What you could see (pause) What you could hear (pause) If you could smell anything (pause). Now I want you to tell me as much as you can about what happened when you came to get your photograph taken.*

With child witnesses, the context of the original event can be recreated by explicitly requesting the child to think about the context and by asking specific questions that require them to think about it. The questions are asked slowly and deliberately with pauses. It is important to emphasise to the child that they must listen carefully to what the interviewer is saying. The instructions to reinstate context were administered as follows in a study where the witnesses were eight and nine year old children being interviewed about a magic show (Memon et al. 1997b):

Interviewer: *Put yourself back to the same place where you saw the magic show. Can you see the room now? (pause) Tell me about the room (pause). Where were you at the*

*time? (pause) Tell me about your feelings (pause). Could you hear anything? (pause) Could you smell anything? (pause).*

### Report Everything

A second technique is to ask the interviewee to *report everything*. This may well facilitate the recall of additional information, perhaps by shifting criteria for reporting information. For instance, witnesses are encouraged to report in full without screening out anything they consider to be irrelevant or for which they have only partial recall (Fisher and Geiselman, 1992). The instruction to ‘report everything’ was used as follows in a study where the witnesses were six and seven year olds being interviewed about an eye test (Memon et al. 1993).

Interviewer: *Just try and tell me what happened, as much as you can remember. If you cannot remember all of it, just tell me what you can. Even little things are important.*

Child: *We had this...even little things?*

Interviewer: *Even little things.*

In addition to facilitating the recall of additional information, this technique may yield information that may be valuable in putting together details from different witnesses to the same crime (see Memon and Bull, 1991). An eyewitness who provides more details is also judged to be more credible in the courtroom (Bell and Loftus, 1989), although the overall accuracy of these details rather than amount of information that is reported should

be the major question. Accuracy of details is of great importance in cases where eyewitness evidence is the main source of evidence that is used to incriminate a suspect (Scheck, Neufeld & Dwyer, 2000). Recent research suggests that the report everything instruction in combination with context reinstatement, might be the most effective component of the CI (Milne & Bull, 2002).

### Change Perspective

A third technique is to ask for recall from a variety of perspectives. This technique tries to encourage the witnesses to place themselves in the shoes of the victim (if the witness is not a victim) or of another witness and to report what they saw or would have seen. Again the aim is to increase the *amount* of detail elicited. Geiselman et al. (1990) report that changing perspectives can be particularly helpful for children if the following instructions are given: 'Put yourself in that other person's body and describe what they would have seen.' However, there are several concerns about the use of the change perspective instruction, in particular the possibility that it could lead to fabricated details and confuse the witness as illustrated by the following example taken from the Memon et al. (1993) child witness interviews about an eye test (see also Boon & Noon, 1994, Roberts & Higham, 2002).

Interviewer: *What I'd like you to try and do is imagine that you are the nurse and that you can see the room from where she was standing, by the wall chart. Just tell me what you can see.*



Child: *Umm...Did you see the letters, can you see the letters good and I said yes and that's all she said to me.*

Evidence obtained using this particular technique may not be easily accepted in legal procedures where it is likely to be seen as encouraging speculation (see Memon and Koehnken, 1992). Moreover, police officers who use the CI have tended not to use the change perspective instruction and have concerns that a witness could be misled when this instruction is applied (Kebbell and Wagstaff, 1996). There is also evidence that the instruction does not increase recall over and above an instruction to 'try harder' (Milne and Bull, 2002).

### Reverse Order

The fourth technique instructs interviewees to make multiple retrieval attempts from different starting points. Interviewees generally feel they have to start at the beginning and are usually asked to do so. However the CI encourages extra focused and extensive retrieval by encouraging witnesses to recall in a variety of orders from the end, or from the middle or from the most memorable event. Geiselman and Callot (1990) found that it was more effective to recall in forward order once followed by reverse order than to make two attempts to recall from the beginning. There is some doubt about whether young children can effectively use this technique as illustrated by the following example from Memon et al, 1993.

Interviewer: *OK. What we are going to do now is tell the whole story backwards. Now the very last thing that you did is you went back up to your classroom.*

Child: *Well, I just walked back and nothing happened.*

Interviewer: *So what happened before you left the room to go back to your classroom?*

Child: *I'm not quite sure*

### **Revisions Of The CI: Improving Communication And Retrieval**

The original version of the cognitive interview resulted in substantial gains (35%) in the amount of correct information that was elicited from eyewitnesses without any apparent increases in errors (e.g. Geiselman et al. 1985). However, in order to be able to effectively implement the use of the 'cognitive' components of the CI, it is necessary to provide interviewers with the necessary social skills and communication strategies that are required in order to build rapport. As indicated earlier, research with police officers suggested this was something they lacked. The revised version of the CI (see Geiselman and Fisher 1997) therefore included the following techniques:

#### Rapport building

This is an attempt to get to know the witness a bit, clarify what the expectations are and generally put the person at ease. An important component of rapport building is for the interviewer to explicitly 'transfer control' by making it clear to interviewees that they are

the ones in control because they are the ones having to work at remembering. Part of transferring control is to give the witness time to think and respond.

### Focused Retrieval

The interviewer facilitates eyewitnesses using *focused* memory techniques (to concentrate on mental images of the various parts of the event such as the suspect's face and use these images to guide recall). Fisher and Geiselman (1992) draw a distinction between conceptual image codes (an image stored as a concept or dictionary definition) and pictorial codes (the mental representation of an image). The notion is that images create dual codes or more meaningful elaborations (Paivio, 1971). The 'imaging' part of the CI usually occurs in the questioning phase of the interview and assumes that the witness has effectively recreated the context in which an event occurred. The instruction could take the following form: 'concentrate on the picture you have in your mind of the suspect, focus on the face and describe it.' In order to effectively engage the interviewee in focused retrieval, the interviewer needs to speak slowly and clearly, pausing at appropriate points to allow the interviewee time to create an image and respond.

### Interviewee Compatible Questioning

Finally the timing of the interviewer's questions is critical (deemed interviewee compatible questioning). Questions should be guided by the interviewee's pattern of recall rather than adhering to a rigid protocol. For example, if an interviewee is describing a

suspect's clothing the interviewer should not switch the line of questioning to the actions of the suspect.

### **Testing The CI: An Effective Control Group**

From a practical perspective, it is important to show that the CI is more effective than the techniques currently in use by police officers and others. From a theoretical perspective, an experimental control is needed to demonstrate that the techniques themselves are causing the effects and not some aspect related to training such as motivation, quality of questioning or rapport building skills. The use of the term 'standard' in earlier studies itself implies inferiority. The issue of experimental control has become even more important in testing the revised version of the CI given that it combines the cognitive techniques with some general strategies for improving communication. What evidence is there that the gains in information elicited are not merely due to the improved communication?

More recent tests of the CI have addressed this question by comparing the CI with a procedure known as the structured interview (SI) where the quality of training in communication and questioning techniques is comparable to the CI procedure. The training of the structured group follows a procedure that is recommended to professionals who interview children (Home Office 2003, Scottish Executive, 2003). The essence of the Home Office guidance is to treat the interview as a procedure in which a variety of interviewing techniques are deployed in relatively discrete phases proceeding from

general to open, to specific, closed-form questions. Rapport-building, through open questions and active listening, is also an important component.

Guenter Koehnken and colleagues (Koehnken et al. 1994; Koehnken et al. 1995; Mantwill et al. 1995) were the first to demonstrate the superiority of the CI as compared with a structured interview. The structured interview group in their studies received training of comparable quality and length to the CI group in basic communication skills. This included instruction on rapport building and use of various types of questioning. The training of the cognitive group also involved the use of the various cognitive techniques. The cognitive and structured interviewers received a sample of written questions and interview transcripts and the trainers critically evaluated the questions and retrieval aids used by the interviewer (CI group). The SIU group focused on the questioning strategy and not the retrieval aids. The interviewers then watched a videotaped interview. Finally each interviewer took part in a role play and received feedback on their performance. The training session lasted between 4-5 hours.

In the Koehnken et al, (1994) study the participants (interviewees) and interviewers were non-psychology students without any prior experience in investigative interviewing. The to-be-remembered event was a videotape showing a blood donation. Participants were tested a week after viewing the event. Each interviewer conducted one interview (n=30). The CI significantly increased the amount of correctly recalled information over the structured interview without increasing the number of errors and confabulated (made-up) details. The interviewers' memory was also tested by asking them to provide, from their

memory, written accounts of the event. The superiority of the CI was noted in the interviewer accounts with an average of 42% more correct details as compared to the structured interview group.

Koehnken et al. (1995) conducted a similar study. The interviews were conducted by psychology students and adults served as interviewees. Again there was an increase in correct details with the enhanced cognitive interview, although this time a small increase in confabulated details was also noted. However, the overall accuracy (proportion of correct details relative to the total number of details reported) was almost identical in both interview conditions (see also Koehnken et al, 1999).

In the Mantwill et al. (1995) study a sample of 58 interviewees was drawn from a variety of professional groups and were selected on the basis of their having experience of a blood donation (the to-be-remembered event) or having no experience. The interviewers were non-psychologists of various professions, none of whom had knowledge of blood donation. The number of details elicited in the cognitive and structured interview were compared. The enhanced CI yielded 25% more correct information without any differences in errors and confabulations. The performance of experienced and inexperienced participants did not differ significantly.

So far the evidence suggests that the enhanced CI effects are due to the use of the cognitive techniques rather than merely a result of enhanced communication. However, this does not fit with the results of a series of studies conducted in England as part of a

project funded by the Economic and Social Research Council in the UK (Memon et al, 1996, 1997b, 1997c). Memon and colleagues directly examined whether the source of the CI advantage was due to facilitated communication arising from the use of the ‘communication’ components of the enhanced CI (namely rapport building and transfer of control) or a result of the ‘cognitive’ components of the enhanced CI (context, imagery, reverse order and reporting in detail). The structured interview in the Memon et al. studies therefore resembled the enhanced CI more closely than the structured interview used in the Koehnken studies due to a greater emphasis on ‘transfer of control.’ As in the Koehnken research, cognitive and structured interviewers received a similar quality of training. The interviewers were psychology students who had no previous experience of interviewing. Both groups were led to believe they were using the superior interview technique. They were trained separately over a period of two days in basic communication techniques such as building rapport, in types of questions to ask and had the opportunity to conduct role-plays and receive feedback on practice interviews. Finally, Memon et al. introduced a ‘second retrieval (SR) phase’ in their CI/SI interviews. In the SR phase of the CI interview, participants were instructed to go through the event again in reverse order. The SI group were simply asked to try one more time. The purpose of this SR was to directly test the hypothesis that the various CI techniques merely increase the number of recall attempts and this accounts for the gains in new information (reminiscence), a well documented effect (see Payne, 1987 for a review). These effects could be a result of retrieval practice (Roediger and Thorpe, 1978) or stimulus sampling whereby repeated attempts to access a memory will result in different samples drawn from a population of encoded details.

The studies conducted by Memon and colleagues showed that the cognitive interview is a highly complex procedure and that the advantage gained from the CI may be restricted to a rather narrow set of conditions. When these conditions are violated the CI appears no better than an effectively matched control group (Memon and Stevenage, 1996). This will be illustrated by considering the results obtained in the Memon et al. research in some detail.

In the Memon et al. (1997b) study where the interviewees were 8 and 9-year-old children, the CI was found to significantly increase correct and incorrect (errors) about a live event (interview 2 days after the event) while accuracy was unaffected. More recent studies using young adults and older adults have also reported that the CI increases correct details as well as incorrect details (e.g. Mello and Fisher, 1996). One interpretation of the increase in errors with the CI is that the use of specific instructions such as the 'report everything' are affecting report criterion (Memon and Higham, 1999). A similar argument has been made about techniques such as hypnosis where it has been argued the use of a more liberal response criterion is likely to increase errors (see Roberts and Higham, 2002). One way to monitor this source of error is to collect confidence ratings from participants (see Roberts and Higham, 2002 for further details). The practical implications of an increase in errors will be considered later in this chapter.

Memon et al. (1997b) compared the cognitive and structured interview as described above with an untrained control group. The interviewers were graduate students and the



interviewees (college students) viewed a videotaped event of a sequence in which a child is murdered. In this study, the cognitive interviewers and structured interviewers did not differ in terms of the overall amount recalled, the accuracy of recall, the number of errors or the number of confabulations (made-up details). Both trained groups elicited more in the way of overall information and amount of correct information relative to an untrained group. However, this was offset by their producing a significantly higher number of errors and confabulations than the untrained group. These findings are important in themselves but they also raise the question of what is the appropriate control group. Clearly, if the cognitive interviewers are compared to an untrained group then they show some advantages. However, if the cognitive interviewers are compared to a group matched for everything but the cognitive techniques (namely the structured interview) then the advantage of the CI disappears.

This finding is at odds with the studies conducted in Germany, and suggests that the effects of the cognitive interview depend on the standard of comparison that is used. It is not clear where there is a discrepancy but it could be due to sampling differences and interviewer differences (Memon's interviewers were psychology graduates and were possibly more motivated). It should be pointed out that while the structured interviewers in the Memon et al. (1997c) study were behaving like cognitive interviewers they did not use any of the cognitive techniques. Of course this does not rule out the possibility that the interviewers were spontaneously using CI techniques. Memon et al. (1997a) found that college students frequently report the use of context reinstatement and imagery as aids to recall. Indeed such techniques are also reported to be used by older participants

drawn from the general population (Harris, 1980). In a recent study of the role of visual imagery in the enhanced CI Davis, McMahon and Greenwood (2004) found that Vividness of Visual Imagery (VVIQ) scores were unrelated to the amount of information correctly recalled. The authors concluded that the rapport building components of the CI and the Structured Interview may evoke spontaneous imagery which is comparable to the explicit instruction to image in a CI. This supports the finding of Memon et al. (1997c).

It appears from the work of Memon and colleagues that the primary effect of the CI is that it enhances communication. The 'transfer of control' instruction could be the most effective component of the CI. Transfer of control is achieved in three main ways. Firstly, the interviewers explicitly tell the interviewee that they do not have knowledge of the event and that it is the interviewee that holds all the relevant information. This confers power and status to the interviewee. Secondly, the interviewers ensure that the interviewee is not interrupted when speaking and pause after interviewees have spoken in case they have more to add. This also conveys to the interviewee that what *they* have to say is important. Finally, whenever possible the interviewers use open ended questions so that it is the interviewee who does most of the talking. The effect of transfer of control may not only be improved rapport but also a more fruitful memory search (e.g. due to interviewers not interrupting) and possibly more effective use of context reinstatement. This explanation could account for the lack of a significant difference between CI and SI group in the Memon et al. (1997c) study.

It should be noted that in the Memon et al. (1997c) study and in earlier studies (e.g. Memon et al, 1997b) there was considerable variability in interviewer performance. The

untrained interviewers in particular adopted quite different styles of questioning, they varied in how persistent they were and how frequently they interrupted interviewees. This has to be taken into consideration in interpreting the results and suggests future studies should use more representative samples and collect some qualitative data on how interviewer background and motivation may influence their performance (see also Memon et al. 1994; Memon, Bull & Smith, 1995).

### Field tests of the CI

To date there have been only two field tests of the Cognitive Interview. The first was a project that enlisted the assistance of police detectives in Miami, Florida (Fisher et al, 1989). The second was a study involving the Hertfordshire police in the U.K. (George, 1991, Clifford & George, 1996).

The aim of the Fisher et al. (1989) field study was to examine the use of the enhanced CI by trained and 'control' police detectives when questioning real life victims and interviewees. The pre-training phase of the study involved the collection of tape recordings of interviews from a sample of detectives using their usual, standard techniques. Half of the group underwent enhanced CI training over four 60-minute sessions. During this time they were given an overview of the procedure and the general psychological principles of cognition, training in specific interviewing techniques, communication techniques and advice on the temporal sequencing of the CI. After the fourth training session each detective tape recorded a practice interview in the field and

received individual feedback from the psychologists on the quality of the interview. The detectives then followed the enhanced CI procedure or 'standard interview' procedure (as defined earlier) during the course of their interviews with real interviewees over a period of time. Two measures of the effects of training were taken. Firstly, number of facts elicited before training (i.e. the tapes from the pilot phase) versus after training (thus a within-participants comparison). Secondly, facts elicited by trained and untrained officers after some had undergone the training programme (between-participants). The tapes were transcribed by trained research assistants, who recorded all the relevant details. Opinionated and irrelevant statements were ignored. The statements included physical descriptions, actions and clothing.

The CI was found to be effective in the before/ after comparisons (the within participants factor) and in the trained/untrained (between participants) comparison. The trained detectives elicited 47% more information after training and significantly more information than detectives not trained in CI. Baseline measures showed there were no differences between the groups prior to training. In order to examine the impact of CI on the amount of incorrect information recalled, it was necessary to examine corroboration rates. (In real crimes, of course, corroborating information is not always easily available). When the corroborating source was another interviewee/ victim, the corroborating interview was always conducted by someone other than the original detective (usually a uniformed officer). Some 94% of the statements from the interviews in this study were corroborated and there was no difference in the corroboration rates of pre- and post-training interviews.

The results of the CI field study are promising. Six of the seven detectives who were trained improved significantly. However, given the relatively small sample size the question as to how representative the trained group were is questionable. The officers were selected for training rather than being randomly assigned to conditions and this is of some concern especially in light of evidence that police officers may not be so open to the use of new techniques (Memon et al, 1994b; 1995). Finally, there was no trained control group in this field study and data is not provided on the techniques used by interviewers (see Memon and Stevenage, 1996)

Clifford and George (1991; 1996) detail a field evaluation of training in three methods of investigative interviewing. The study involved 28 experienced British police officers, who were randomly assigned to one of four conditions: CI; conversation management; CI and conversation management; and, a no training control group. Conversation management is a procedure that resembles the structured interview described earlier and includes training in planning the interview, listening skills, conversational styles, question types and summarising. Prior to training each participant provided a tape recording of an interview they had conducted with a real life interviewee or victim. Following training each police officer tape-recorded three more interviews with victims or interviewees of street crimes. The tape recordings were transcribed and evaluated, among other things, for the amount of information provided by the interviewee. The results showed that the CI elicited significantly more information than the standard police interview (14% more than the no training control group). A before and after training

comparison showed an increase of 55%. There were no significant differences between conversation management and the untrained control, in fact the conversation management group fared worse.

At this point it is necessary to point out two things. Firstly the success of the CI depends upon adequate training of interviewers in the techniques described above. It is not clear how much training is needed. Some studies report effects with relatively brief training. Fisher et al. (1989) report benefits after four 60-minute sessions, while George (1991) trained officers over 2 days. Memon et al, (1994b) trained officers over a more limited period of time (4 hours) and found this was insufficient to motivate officers to use the new techniques while Turtle (1995) has evaluated several one week training courses on the CI for Canadian police officers and found that training has relatively little effect on the use of CI techniques. Clearly, the effects of training are complex and depend not only on length of training, but quality of training, background of interviewer, attitudes towards training and so forth. A survey of police officers (Kebbell, Milne & Wagstaff, 1999) suggests relatively few officers use the full CI in practice and one reason for this may be that they do not feel they are sufficiently trained in its use.

Finally, it should be noted that the CI relies upon a co-operative interviewee. It has not as yet been established how useful a CI would be in the case of an interviewee who does not wish to remember (e.g. in the case of traumatised interviewee) or an interviewee who does not wish to communicate information to the interviewer (e.g. one that is likely to be a suspect). There is also no evidence that the CI is a reliable procedure for eliciting memories from adult clients of childhood events (Memon, 1995). It is therefore of some

concern that the CI is mentioned in the British Psychological Society guidelines on good practice when working with recovered memories (Frankland & Cohen, 1999).

So far the review of evidence has focused largely on studies using adult interviewees (with the exception of the Memon et al, ESRC research) one important question is to what extent can the CI be effectively used with child interviewees?

### The CI and Child Interviewees

Research has consistently shown that younger children (ages 6-7) will often recall *less* information than older children (ages 10-11) (see Memon et al, 2003; Westcott, Davies and Bull, 2002 for reviews). Given that the primary aim of CI is to increase the *amount* of information retrieved it may be a most effective procedure to use with young children.

In one of the earliest studies of the CI using children, Saywitz et al. (1992) refined the procedure for children (7-8 year olds and 10-11 year olds) using a live event. The to-be-remembered event was a game where children dressed up, interacted with, and were photographed by a stranger. The CI significantly increased correct facts recalled across both age groups by 26%. In a second study, a 'practice session' was included, the aims of which were to familiarise the children with the cognitive interview techniques and to give feedback on their performance. The interviewers were experienced police officers who received written instructions and a two hour training session during which they were

informed about child appropriate language, rapport building, interview preparation and procedure. The CI group received additional information on the use of the four original CI techniques. The standard interview group were instructed to use the techniques they would normally use. The CI led to more correct details being recalled than did the SI (20% increase for the 8-9 year olds and 44% for the 11-12 year olds). Furthermore, collapsing across age levels, a practice cognitive interview prior to the main interview improved performance by an additional 25%. No increase in the amount of incorrect or confabulated details was observed. Thus it may help to familiarise children with the CI before they are questioned about the event. Interestingly it was the older children whose performance improved most with a practice interview which is the opposite to what we would expect based on our understanding of children's development of memory strategy usage (Ornstein et al. 1985). An analysis was conducted to look at the frequency with which interviewers used the four CI techniques. It was apparent that the student interviewers were more likely to use each of the four techniques compared to the experienced detectives. For example only half of the detectives used the 'change perspective' instruction as compared to 94% of the student interviewers. George (1991) in the British field study and Turtle (1995) also found police officers did not use this technique even though they had been trained to do so.

The Saywitz et al. study provides a powerful demonstration of the effectiveness of a CI with children over the age of 8 years. There are however several concerns. Firstly, the CI groups received training while the standard group did not. This may affect the motivation of the CI interviewers (Memon et al., 1994b). Secondly while the authors of



this study did attempt to correlate use of each CI technique with overall memory scores we have no information about the types of details elicited with the various CI techniques. A related point is that the details about errors are not provided. Thirdly, while the CI practice group had an opportunity to ‘practice and become familiar with the task of retrieval, the control group did not. In more recent research Koehnken et al. (1992) investigated the effectiveness of the enhanced CI with 51 nine and 10 year old children who had been shown a short film. After a delay of three to five days, they were questioned by trained psychology students about the film using either an enhanced CI or SI (structured) interview. In this study great care was taken to ensure that the SI interviewers were trained in the same way as CI interviewers save for using the special CI techniques. The enhanced CI produced a 93% increase in the amount of correct information recalled compared to the SI. The number of confabulations (the reporting of details not present in the event) increased, however, and this has been found in several studies where the CI had been used with child interviewees (e.g. Mantwill, et al, 1995).

Memon et al. (1993) interviewed 32 six and seven year olds about an eye test. The effectiveness of a CI, comprising the usual four mnemonic techniques was compared with that of a structured interview. The latter, like the CI, was a good interview procedure and one in which the interviewers used open ended questions and did not interrupt the interviewee when he or she was speaking. Children’s recall of the event was tested one week after the event and again six weeks later. As illustrated earlier, children had difficulty in understanding interviewer instructions to change perspective and to recall in reverse order such that they became confused about what was expected of them.

This may have worked against the CI. Indeed there were no significant differences in the types of information elicited between each type of interview with the exception of information about locations of objects and people, which was significantly greater with the CI. This increase in location information was possibly a result of the language used to fulfil the context instructions (e.g. 'describe the room') rather than a direct product of memory improvement (one of the problems of the CI research has been the difficulty separating the two).

Memon and colleagues have attempted to isolate the effects of the individual components of the original CI (Memon et al, 1994a) and have tested the effectiveness of the CI with younger children (aged 6-7 years). Each of the four mnemonic techniques of the CI (context, report in detail, change order and change perspective) was compared with an instruction to 'try harder'. Prior to each interview there was a practice session in which each child described a familiar activity using one of the four CI techniques e.g. context reinstatement. There were no significant differences in correct recall or errors as a function of instruction condition, this suggested that the 'try harder' instruction could be as effective as each of the CI techniques. There were a number of interesting differences between the age groups, most notably that the younger children (5 year olds) performed less well under the CI 'context reinstatement' and 'change perspective' conditions as compared to the 8 year olds in the same conditions. A qualitative analysis of the interview transcripts suggested that the children did not fully understand all the techniques and had difficulty using the change perspective instruction. As Ornstein

(1991) points out, a good interviewer should tailor the interview so that it takes into consideration the cognitive and linguistic capabilities of an individual child.

A more recent componential analysis of the CI was undertaken by Milne and Bull (2002).

They examined the relative effectiveness of each of four original components of the CI mnemonics in three age groups: 5-6 year olds, 8-9 year olds and adults. The children were shown a video of an accident and 48 hours later the participants were interviewed with one of the following six techniques: context reinstatement, change order, change perspective, report everything, context and report everything or a control 'try harder' instruction. The only condition which elicited more recall than the control instruction was a combination of context and report everything. Consistent with Memon et al.

(1994a) none of the other CI techniques differed in the amount of recall elicited as compared to the try harder instruction. Surprisingly there was no effect of age group on the amount of recall elicited with the different instructions. In a study on the accuracy and forensic relevance of information obtained with the CI using adults, Roberts & Higham, 2002 found that the mental reinstatement of context and report everything instruction used in the free narrative was associated with the most correct and relevant details. This supports the results of Milne and Bull (2002) with children.

#### Limitations Of The Use Of CI With Child Interviewees

Several problems in applying the CI with children have been identified. Firstly, younger children (6-7 years of age) have difficulty understanding the CI techniques in the form

developed for adults although more recent studies report that even 4 year olds can effectively use a modified version of the CI (see below). Secondly, the CI when used with children can on occasion increase errors (Memon et al, 1995a; Milne et al, 1995) and confabulations (Koehnken et al, 1992). Thirdly, the CI procedure may also increase demand characteristics in that children respond in a way they think may please the interviewer. This is illustrated in the following transcript taken from Memon et al. (1993) child interviewee study.

Interviewer: OK, but what about the day the nurse came, can you tell me about that. I know you've told me already, but I need to find out more.

Child: *Yes, just in case I am saying the right things.*

Interviewer: *Well no, not exactly, but just in case you remember any more.*

Child: *Yes, some different things that I forgot to say at all.*

Interviewer: *Yes. The best thing to do is to start again and tell me everything again.*

Child: *Well, I can't tell you the same things.*

### The Use Of CI With Special Populations

This section briefly considers recent research which has examined whether the CI can aid the recall of vulnerable witnesses namely very young children, senior citizens and adults and children with learning disability.

To date, there have only been two published studies of the efficacy of the CI when the witnesses are older adults. Mello and Fisher (1996) found the CI led to similar increases in correct recall when the participants were older adults (mean age= 72 years) but Searcy, Bartlett, Memon & Swanson (2001) found no differences in correct identification (recognition) of a culprit when witnesses (ages 62-79 years) were interviewed with an interview procedure resembling the CI. The failure to find an effect of CI on recognition (in this case identification of a culprit) is consistent with earlier studies (see Geiselman and Fisher, 1997 for a review). It may be the case that the CI is more effective in the earlier phases of an investigation when a witness is having to retrieve information from memory in the absence of cues.

Holliday (2003a) has recently reported that children as young as 4 and 5 years old are able to use the Cognitive Interview instructions (context reinstatement, report everything and reverse order) and generate more details in free recall when the CI techniques are used. Furthermore, with the same age group Holliday and Albon (in press) report that a CI given after misinformation reduced children's reporting of misinformation at interview (see also Holliday, 2003b). The latter finding is consistent with Memon, Wark, Holley, Bull and Koehnken (1996a) who studied 8-9 year olds, but is at odds with that of Hayes and Delamothe (1997) who found susceptibility to misinformation in children (ages 5-7 and 9-11 years) was unaffected by interview type.

There is some evidence to suggest the CI can aid the recall of adults (Milne, Clare and Bull, 1999) and children (Milne and Bull, 1996) with mild learning disabilities although

further research is required with this population using larger sample sizes and people with a broader range of learning difficulties.

### Interviewer Differences

It is generally accepted that interviewee performance varies with the quality of the interviewer and vice versa. There is evidence that interviewer performance will vary with interviewer expectations and with level of training (see Memon, et al., 1994b; Memon et al. 1995 and Memon et al. 1996b) for details. It is possible that extensive training and practice may iron out some of these differences and also make interviewers more flexible when it comes to adapting the procedure to suit the needs of particular interviewees such as young children.

### The CI And Delay

Witnesses may not be questioned by the police until some time after they have experienced an event and there may be further delays from the first interview to any subsequent interview. It is therefore important to establish whether the benefits of a CI can be seen over a delay longer than a week.

Memon, Wark, Koehnken and Bull (1997b) asked children (aged 8-9 years) to view a magic show and interviewed them after a short delay of 2 days (Time 1) and/or a longer delay of 12 days (Time 2). At Time 1 the CI produced a significantly greater amount of

correct recall than did a structured interview. However, at Time 1 the CI also produced significantly more recall errors. At Time 2, no differences occurred between CI and SI recall. There was however, a significant hypermnesia effect (a net increase in amount recalled over time) and this might account for why there was no CI effect at Time 2.

Larsson, Granhag and Spjut (2003) were interested in whether the positive effects of the Cognitive Interview hold over time (six months). In their study, 10 and 11 year olds viewed a film and were either interviewed 7 days or 6 months later with a CI or a Structured Interview. Larsson et al. found that children in the CI condition recalled significantly more correct details at both retention intervals although children interviewed after 7 days were more accurate.

Akehurst, Milne and Koehnken (2003) examined whether the enhanced cognitive interview would aid recall with children ages 8-9 years and 11-12 years when there was a 6-day delay. Children viewed a video of a shoplifting incident and were interviewed four hours or 6 days later. The CI led to an increase in correct recall with no increases in errors. There were no interactions involving age group or delay.

## **Overview**

The cognitive interview (CI) emerges as probably the most exciting development in the field of eyewitness testimony in the last twenty years. It presents itself as a technique to facilitate recall, and initial tests show consistent gains in the amount of information that

can be gathered from an interviewee with a cognitive interview. Over the years, studies of the CI have improved in design in that they have shown the CI can yield more correct information than interviews that contain the same structure and rapport building. The only concern that remains is that the CI can occasionally increase errors particularly when the witnesses are young. Obviously, the benefits of any innovative technique need to be carefully weighed up against any costs. In a forensic investigation an increase in number of details could be especially helpful at the information gathering stage in providing new clues that could lead to a successful conviction. On the other hand, what if investigators are led up the wrong path? Thus research which helps us understand precisely when and why the cognitive interview will be effective and yield forensically relevant information is valuable (see Roberts & Higham, 2002 for a recent example).

Until recently, the recommendation was that the Cognitive Interview should only be used with older children (over the ages of 7 years). However, recent research suggests developmentally modified versions of the CI can elicit twice as many correct details as control interviews in 4 and 5-year-olds (Holliday & Albon, in press).

Further research is required on how the quality of the interviewer and effectiveness of CI training impacts on the quality of evidence. Training should be monitored over a reasonable period of time in order to assess the effects of feedback and experience in use of the CI techniques on interviewer performance. Furthermore, it is essential that researchers consider the effects of interviewer variables either by including interviewers as factors in the design of experimenters and/or by collecting baseline data on interviewer



performance prior to training. Practitioners are advised that a full training programme in Cognitive Interview together with follow-up training sessions be devised before the techniques are used in the field.

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